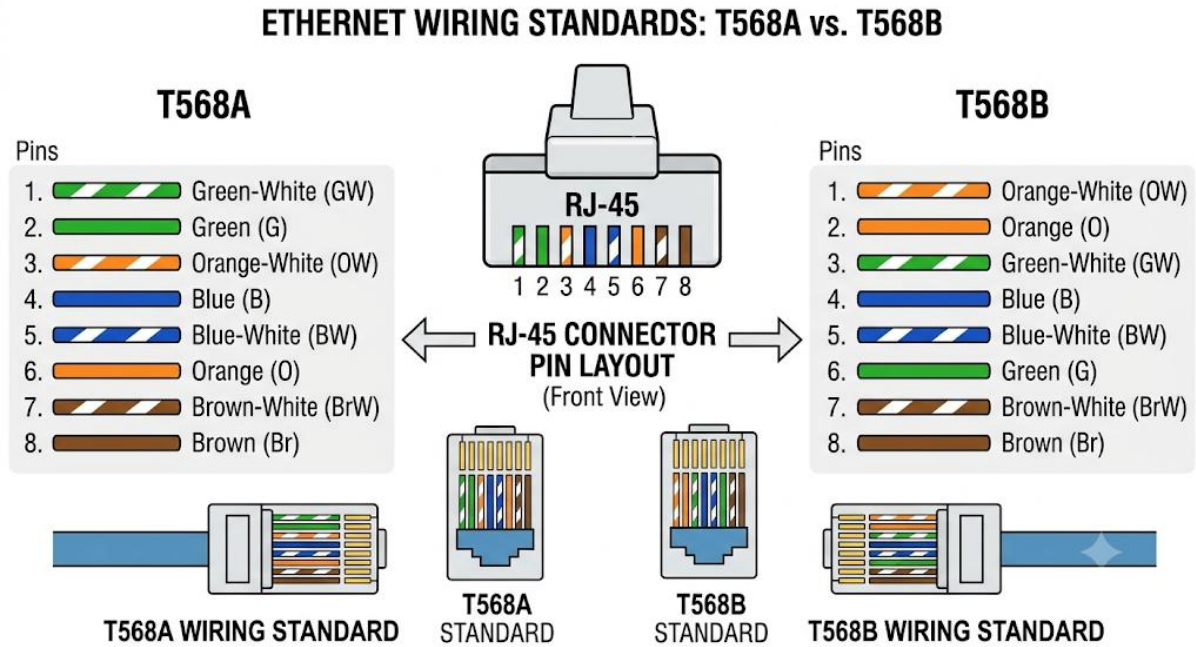


Ethernet standard and specification

Task (1):



Task (2):

Comparison of Ethernet Standards

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1 Gbps (1000 Mbps)
Cable Type	Cat3 or better twisted-pair	Cat5 or better twisted-pair	Cat5e or better twisted-pair
Maximum Cable Length	100 m	100 m	100 m
Connector Type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Common Use	Older/basic Ethernet networks	Fast Ethernet networks	Gigabit Ethernet, modern LANs

Task (3):

Directly Connecting Two Laptops

For connecting two laptops directly using Ethernet, traditionally you would use a crossover cable.

- Cable: Ethernet crossover cable
- Connector: RJ-45
- Why: A crossover cable swaps the transmit (TX) and receive (RX) pairs, allowing two similar devices, such as two laptops, to communicate directly without a network switch or router.

Task (4):

Three Real-World Devices Using Ethernet:

Device	Likely Ethernet Standard	Reason
Wi-Fi Router	1000BASE-T (Gigabit Ethernet)	Modern routers commonly provide Gigabit LAN ports for fast Internet and local network connections.
Desktop Computer / College PC	1000BASE-T	Gigabit Ethernet provides fast and reliable connections for Internet access, file sharing, and online applications.
Network Printer	100BASE-TX or 1000BASE-T	A wired Ethernet connection allows multiple computers to reliably share the printer over the network.